R Practice 2 — Data Structures

This assignment is just for your own practice and does not need to be submitted. If you have any questions, please feel free to reach out to me, come to office hours, or schedule another a time to meet with me.

Open an R Script in RStudio and complete each of the following:

1. **Vectors**
    a. Create a numeric vector named **temperatures** containing the daily temperatures (in degrees Celsius) for a week: 25, 28, 30, 27, 23, 22, 26. Calculate and print the average temperature for the week.

    b. Create a character vector named **days** containing the names of the days of the week: "Monday", "Tuesday", ..., "Sunday". Print the vector.

    c. Create a numeric vector named **test_scores** containing scores of 10 students: 85, 92, 78, 60, 95, 88, 68, 82, 91, 75. Calculate and print the mean, median, standard deviation, and IQR of the test scores.

    d. Calculate and print the mean, median, standard deviation, and IQR for test scores 1, 2, 3, and 10 (all four of those together). Use square brackets to subset the vector.

    e. Create a logical vector (that just contains TRUE or FALSE) named **passing** based on whether each student's test score is 70 or above (considered passing). Calculate the percentage of students who passed the test. You can do this with the **mean()** function.

    f. Calculate and print the mean, median, standard deviation, and IQR of the test scores for only those students that passed the test (got a 70 or above). You can do this using **test_scores[passing]** after you define the passing vector in part e and putting that into the functions. Or you can do **test_scores[test_scores >= 70]**.

2. **Data Frames**
    a. Create a data frame named "students" with the following columns:
        - "Name": "Alice", "Bob", "Catherine"
        - "Age": 25, 22, 23
        - "Grade": "A", "B", "A-"

    b. Add a new column named "Sex" to the "students" data frame with the values "Female", "Male", "Female". Print the updated data frame.

3. **More Data Frames**
    a. Copy the following code into **R**.

```
DF <- data.frame(x = c(10, 12, 8, 8, 7, 10, 6, 15, 16, 20, 12, 21),
          y = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))
```

    b. Change the name of the "x" variable to "sales" and change the name of the "y" variable to "month".

    c. Add a new column to the data frame called "revenue" that is defined by $revenue = 100 \cdot sales - 80$. Also change the name of the data frame from "DF" to "sales_info". Print the "sales_info" data frame.

    d. Remove the "DF" data frame using the **rm()** function.

    e. Calculate and report the mean of the "revenue" variable two different ways: using the $ operator and using square brackets [].

    f. Calculate and report the mean of the "revenue" variable for the months of June, July, and August.

4. **Even More Data Frames!**
    a. Load the built-in "mtcars" data frame in **R** using the **data()** function, which contains information about various car models. Display the first 10 rows of the data frame.

    b. Filter the data to show only cars with more than 100 horsepower and sort them in descending order of horsepower (largest horsepower is first). You can do this using the **order()** and the **which()** functions. Print the filtered data.

5. **Factors**
    a. Create a character vector called "grade_level" with 6 freshmen, 2 sophomores, 3 juniors, and 4 seniors.

    b. Create a grade level factor called "grade_factor" using the **factor()** function. Make sure the levels of this factor are in the order freshmen, sophomores, juniors, then seniors. By default, the factor levels will be alphabetical, which we do not want here